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ORIGINAL ARTICLE

Efficacy and Safety of Finerenone in Patients With Primary Aldosteronism: A Multicenter Prospective Study

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BACKGROUND: Finerenone is a novel nonsteroidal mineralocorticoid receptor antagonist. However, robust evidence about its efficacy and safety in primary aldosteronism is scarce.

METHODS: In this prospective, multicenter, single-arm, and exploratory trial, we enrolled adults (aged ≤75 years) with primary aldosteronism, an office blood pressure (BP) ranging from 140 to 180/90 to 120 mm Hg, and an estimated glomerular filtration rate ≥60 mL/min per 1.73 m². Eligible patients received finerenone (20–40 mg/d) treatment for 12 weeks. The primary outcome was the change in daytime systolic BP at 12 weeks.

RESULTS: Fifty-seven patients were ultimately treated. Per-protocol analysis revealed that finerenone treatment significantly reduced mean daytime systolic BP (−6.69±1.60 mm Hg; $P<0.001$) and diastolic BP (−4.55±1.06 mm Hg; $P<0.001$) according to ambulatory monitoring. Mean office BP decreased even more substantially (systolic BP, −15.58±1.69 mm Hg; diastolic BP, −8.61±1.02 mm Hg; both $P<0.001$). The mean increase in serum potassium concentration was 0.39±0.05 mmol/L, and 94.5% of patients exhibited a normal concentration after 12 weeks of treatment (versus baseline 61.8%; $P<0.001$). Plasma renin activity increased, and 32.7% of patients exhibited a plasma renin activity concentration ≥1 ng/mL per h. According to the Primary Aldosteronism Medical Treatment Outcome criteria, 29.1% and 20.0% of patients achieved complete biochemical and clinical responses, respectively. Treatment was well tolerated.

CONCLUSIONS: This study demonstrated the efficacy and safety of finerenone in the treatment of primary aldosteronism, supporting its use as a potential alternative therapy for the condition. Nevertheless, further prospective and head-to-head randomized controlled trials are essential to establish finerenone as a viable substitute for spironolactone.

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Key Words: antihypertensive agents ■ blood pressure monitoring, ambulatory ■ hypertension ■ mineralocorticoid receptor antagonists ■ renal insufficiency, chronic

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As the most prevalent cause of secondary hypertension, primary aldosteronism (PA) accounts for 4% to 21.6% of hypertensive cases in recent epidemiological studies,[1–3] and its prevalence increases to 11.3% to 29.1% among populations with resistant hypertension.[3–5] Compared with essential hypertension, PA is associated with a greater risk of cardiovascular and cerebrovascular complications, metabolic diseases, and renal

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NOVELTY AND RELEVANCE

What Is New?

This initial multicenter prospective study demonstrates the efficacy and safety of finerenone in patients with primary aldosteronism and provides the first evidence to support a comprehensive clinical and biochemical assessment of finerenone in the context of treating primary aldosteronism.

What Is Relevant?

Finerenone treatment significantly reduced both office blood pressure and 24-hour ambulatory blood pressure (more notably for nocturnal blood pressure) while increasing serum potassium and plasma renin activity.

Clinical/Pathophysiological Implications?

Our findings identify finerenone as a potential alternative therapy for primary aldosteronism, particularly for patients who are intolerant to the adverse effects of spironolactone.

Nonstandard Abbreviations and Acronyms

ABPM	ambulatory blood pressure monitoring
AE	adverse event
ASI	aldosterone synthase inhibitor
BP	blood pressure
CKD	chronic kidney disease
DBP	diastolic blood pressure
eGFR	estimated glomerular filtration rate
MRA	mineralocorticoid receptor antagonist
PA	primary aldosteronism
PAMO	Primary Aldosteronism Medical Treatment Outcome
PRA	plasma renin activity
SBP	systolic blood pressure
UACR	urinary albumin-to-creatinine ratio

dysfunction.[6–8] In addition, PA significantly increases all-cause mortality rates.[9] Thus, early diagnosis and precision treatment are of paramount importance.

Current therapeutic strategies for PA are determined by its etiological classification. Unilateral adrenal hypersecretion, which occurs in ≈30% of cases, may be treated with adrenalectomy. However, the majority of patients (≈60%) exhibit bilateral adrenal involvement, which requires lifelong medical therapy.[10] Currently, mineralocorticoid receptor antagonists (MRAs) are the main treatments. Spironolactone is widely used owing to its high efficacy, whereas its low-selective binding to androgen and progesterone receptors frequently leads to dose-dependent adverse effects on the gonads.[11,12] Eplerenone, although more selective, demonstrates suboptimal antihypertensive efficacy and carries substantial cost burdens.[13] Emerging aldosterone synthase inhibitors (ASIs) will play a role in the medical management of PA. However, to date, only phase 2 evidence is available.[14] These limitations highlight the need for better therapeutic alternatives for PA.[15,16]

Finerenone, a novel nonsteroidal MRA, exhibits superior affinity and selectivity for the mineralocorticoid receptor.[17] In addition, compared with steroidal MRAs, the considerable nonsteroidal scaffold and distinct physicochemical properties of finerenone confer differences in mineralocorticoid receptor antagonistic potency, coregulator recruitment, tissue penetration, and distribution profile.[17] Large-scale randomized clinical trials have shown that finerenone effectively reduces cardiovascular and renal risks in patients with chronic kidney disease (CKD) and type 2 diabetes.[18–20] To date, clinical data evaluating the efficacy of finerenone for PA are scarce, with only 2 published studies. One single-center pilot study with a relatively short duration demonstrated that finerenone and low-dose spironolactone had comparable antihypertensive effects in patients with PA.[21] Another real-world study with a small sample (n=15) indicated that although finerenone achieved blood pressure (BP) similar to that achieved with eplerenone, there was a lower proportion of patients with normal BP and complete biochemical response.[22] Therefore, it is essential to clarify the efficacy and safety of finerenone in the treatment of PA through multicenter studies. Moreover, newly published criteria for evaluating the therapeutic efficacy of medical treatments in PA enable a more comprehensive and objective assessment of pharmacotherapy outcomes.[15] Currently, no evidence exists about the comprehensive clinical and biochemical evaluation of finerenone for PA.

We conducted a multicenter, prospective, single-arm, exploratory clinical study to evaluate the efficacy and safety of finerenone in patients with PA by thoroughly assessing clinical and biochemical outcomes following medical treatment.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Study Design

This multicenter, prospective, single-arm clinical trial was conducted across 8 hospitals in China. A list of study sites and investigators is provided in Table S1. This study aimed to assess the efficacy and safety of finerenone in patients with PA.

Eligible patients were enrolled in a run-in period and discontinued prior antihypertensive agents. These patients received a stable dose of nifedipine controlled-release tablets (0–60 mg/d) as needed for BP management from the 2-week run-in period until the end of the treatment period. The run-in period was extended to 4 weeks in patients who previously took MRAs. For patients with serum potassium levels of <3.0 mmol/L, oral potassium supplementation was administered and was discontinued 2 weeks after baseline. Baseline evaluations were performed after the run-in period. Patients who met the inclusion criteria subsequently proceeded to the treatment period and initiated finerenone at 20 mg once daily. The treatment period spanned 12 weeks, with a telephone visit at week 2 to monitor BP and adverse events (AEs). In-person visits occurred at baseline and at weeks 4, 8, and 12 to assess office BP, laboratory findings, and AEs. If BP remained ≥140/90 mm Hg, serum potassium was <5.0 mmol/L, and the estimated glomerular filtration rate (eGFR) decreased by <30% from baseline at the week 4 assessment, the finerenone dose was increased to 30 or 40 mg daily, administered in 2 divided doses. Thereafter, during weeks 4 to 8 of treatment, the doses were adjusted until the BP targets were achieved or the maximum dose was reached with no further adjustments. In addition, 24-hour ambulatory BP monitoring (ABPM) was performed at baseline and at week 12 for all patients. Following treatment completion, no restrictions were imposed on patients' medication use, and a telephone visit was conducted at week 16 to review the medication regimens and BP control.

Patients

The study enrolled patients who were diagnosed with PA and chose pharmacological intervention, who were between the ages of 18 and 75 years, and who had a confirmed history of hypertension. The inclusion criteria specified office systolic BP (SBP) levels ranging from ≥140 to <180 mm Hg and diastolic BP (DBP) levels ranging from ≥90 to <120 mm Hg during the 2 most recent assessments within the observation period along with an eGFR of ≥60 mL/min per 1.73 m^2. The key exclusion criteria were the presence of secondary hypertension, hypertensive crises, orthostatic hypotension, heart failure, severe hepatic impairment, and serum potassium levels <2.5 or ≥5.0 mmol/L. The comprehensive inclusion and exclusion criteria are detailed in Table S2.

BP Measurements

Office BP measurements were obtained by the investigators using an Omron HEM-7124 BP monitor (Omron Healthcare Co Ltd, Kyoto, Japan). After the patient rested for a minimum of 5 minutes in a seated position, 3 successive readings were taken from the arm with higher baseline BP values with at least a 1-minute interval between measurements. The mean of 3 readings was used in the analysis. BP assessments were conducted 1 to 3 hours after medication administration. Furthermore, 24-hour ABPM was performed by experienced clinical personnel. Daytime measurements were taken every 30 minutes, whereas nocturnal readings were obtained hourly. The minimum requirement included a success rate of 70% for the readings (typically observed at ≥27 measurements over 24 hours). Preferably, 7 nocturnal readings should have also been obtained. Consistent use of the same monitoring device was ensured for each participant, both pre- and post-treatment.

Laboratory Assessment

Blood chemistry, urinary microalbumin, and hormone levels were assessed at baseline and at weeks 4, 8, and 12. The parameters included alanine aminotransferase, aspartate aminotransferase, total bilirubin, total cholesterol, triglycerides, high-density lipoprotein, low-density lipoprotein, blood urea nitrogen, creatinine, electrolytes, the urinary albumin-to-creatinine ratio (UACR), plasma renin activity (PRA), and aldosterone. Blood samples were collected from patients after an overnight fast of at least 8 hours and processed immediately. Blood chemistry was analyzed using an autoanalyzer, whereas urinary microalbumin was measured using the nephelometric method. PRA was measured by radioimmunoassay, and aldosterone was quantified using chemiluminescent immunoassay. All laboratory tests were conducted at the local laboratories of the participating sites.

Outcomes

The primary outcome was the change in daytime SBP measured by 24-hour ABPM from baseline to week 12. The secondary outcomes included changes in daytime DBP, 24-hour mean SBP and DBP, nocturnal SBP and DBP, the rate of decline in nocturnal SBP and DBP, and office SBP and DBP. Additional secondary outcomes evaluated the rate of hypertension remission at week 12 (defined as the proportion of patients with office BP <140/90 mm Hg), the proportion of patients with normal serum potassium levels (defined as 3.5–5.5 mmol/L), and changes in serum potassium levels, PRA, aldosterone, eGFR, and UACR. Furthermore, the study evaluated biochemical and clinical response rates according to the Primary Aldosteronism Medical Treatment Outcome (PAMO) criteria.[15] Safety assessment outcomes included the incidence of hyperkalemia, AEs, and serious AEs, which were recorded on the basis of participants' self-reports or routine inquiries and examinations during visits.

Sample Size

This study used a single-arm design to assess the change in daytime SBP from baseline to week 12 as the primary outcome. A meta-analysis suggested that a 10-mm Hg reduction in SBP correlated with a decreased risk of cardiovascular events.[23] Therefore, a mean SBP difference of 10 mm Hg with an SD of 20 was assumed posttreatment.[24] Fifty-five patients were required to detect pretreatment and posttreatment changes using a 2-sided test with 5% alpha, 90% power, and 20% patient attrition. The sample size was determined using PASS software, version 11.0 (NCSS LLC, Kaysville, UT).

Statistical Analysis

The safety analysis set included patients who received at least 1 dose of finerenone and had data that allowed safety assessment. The efficacy analysis set included patients who provided

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informed consent and completed a 12-week finerenone treatment with primary outcome data.

Continuous variables are presented as the means and SDs or medians and interquartile ranges, whereas categorical variables are presented as frequencies and percentages. Paired *t* tests were utilized for normally distributed variables, and Wilcoxon signed-rank tests were used for nonnormally distributed variables to compare efficacy before and after treatment. Changes before and after medication administration are presented as the means with SEs or the medians with interquartile ranges. The statistical outcomes for variables at baseline and week 12, accompanied by their respective *P* values, are summarized in Table 2. All *P* values reported are 2-sided, and values <0.05 were considered statistically significant. A sensitivity analysis of the primary outcome was also conducted on patients who withdrew prematurely. Missing data at week 12 were imputed using baseline values. All statistical analyses were performed using SPSS 26.0 software (IBM Corporation, Armonk, NY).

Ethics

All patients provided written informed consent before enrollment. The study protocol was approved by the local ethics committees or institutional review board at each participating center. The study adhered strictly to institutional guidelines, the Declaration of Helsinki, and good clinical practice standards. This study was registered with ClinicalTrials.gov on April 24, 2024.

RESULTS

Patients

A total of 142 patients underwent screening. Seventy-two patients were ineligible or declined, and 70 patients were subsequently recruited during the run-in period. A total of 57 patients ultimately entered the treatment period. Among the 57 patients who received finerenone therapy, 55 (96.5%) had complete primary outcome data at week 12 (Figure 1).

The study cohort comprised mostly males (70.2%) with an average age of 51.2±10.0 years. The median duration of hypertension was 10.0 (5.0–18.5) years, with the highest recorded office BP of 168.0 (155.5–181.5)/103.0 (100.0–110.0) mmHg. Nine patients (15.8%) had resistant hypertension. Before the run-in period, 45 patients (78.9%) were on calcium channel blocker antihypertensive medication, 11 patients (19.3%) received an angiotensin-converting enzyme inhibitor or angiotensin receptor blocker, and 16 patients (28.1%) received MRAs. Furthermore, 47 patients (82.5%) had a history of hypokalemia, with the lowest recorded serum potassium level of 2.9±0.4 mmol/L. Computed tomography scans revealed adrenal lesions in 51 patients (89.5%), 43.1% of whom had unilateral lesions and 56.9% of whom had bilateral lesions. Adrenal venous sampling was performed on 29 patients (50.9%). One patient had a history of adrenalectomy (Table 1).

During the run-in period and up to week 12, 14 (24.6%) patients did not receive nifedipine as background medication, 31 (54.4%) received 30 mg, and 12 (21.1%) received 60 mg daily, resulting in an average dose of 28.95 mg. After the baseline evaluation, finerenone at a daily dose of 20 mg was introduced. At the end of the study, 13 (23.6%), 9 (16.4%), and 33 (60.0%) patients received daily doses of 20, 30, and 40 mg, respectively, with an average dose of 33.64 mg. Initially, potassium chloride sustained-release tablets were provided as supplements to 10 patients (17.5%) at baseline.

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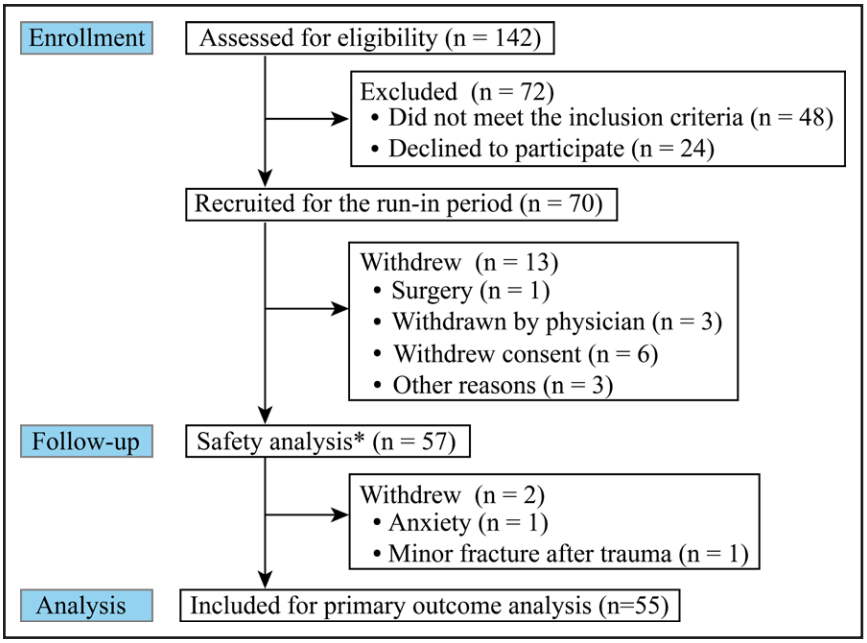

Figure 1. Trial profile.
*Patients eligible for the safety analysis had at least 1 dose of finerenone and data that enabled a safety assessment.

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Table 1. Demographic and Clinical Characteristics of the Patients at Baseline

Characteristics	N=57
Age, y	51.2±10.0
Male, n (%)	40 (70.2)
BMI, kg/m^2	26.5±2.7
Smoker, n (%)	19 (33.3)
Drinker, n (%)	17 (29.8)
Duration of hypertension, y	10.0 (5.0–18.5)
Highest systolic blood pressure, mm Hg	168.0 (155.5–181.5)
Highest diastolic blood pressure, mm Hg	103.0 (100.0–110.0)
Resistant hypertension, n (%)	9 (15.8)
Types of antihypertensive drugs taken before, n (%)	
Calcium channel blockers	45 (78.9)
ACE inhibitors or ARBs	11 (19.3)
β-Blockers	11 (19.3)
MRAs	16 (28.1)
Other	14 (24.6)
History of hypokalemia, n (%)	47 (82.5)
Minimum serum potassium, mmol/L	2.9±0.4
CT findings, n (%)	
Normal	6 (10.5)
Abnormal	51 (89.5)
Bilateral	29 (56.9)
Unilateral	22 (43.1)
Adrenal venous sampling, n (%)	29 (50.9)
Bilateral	21 (72.4)
Unilateral	8 (27.6)
History of surgical treatment, n (%)	1 (1.8)

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BMI, body mass index; CT, computed tomography; and MRA, mineralocorticoid receptor antagonist.

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Efficacy

All efficacy outcomes are summarized in Table 2.

The primary outcome of reduction in daytime SBP from baseline to week 12 was 6.69±1.60 mm Hg (week 12: 131.71±12.09 mm Hg versus baseline: 138.40±12.04 mm Hg; $P<0.001$), and the nocturnal and 24-hour mean SBP decreased by 9.64±1.60 and 7.75±1.48 mm Hg, respectively (both $P<0.001$ versus baseline). The daytime, nocturnal, and 24-hour mean DBPs decreased by 4.55±1.06, 5.38±1.00, and 5.02±0.95 mm Hg, respectively (all $P<0.001$ versus baseline). Notably, the nocturnal SBP decline rate increased significantly from baseline (week 12: 7.14±6.46% versus baseline: 4.99±6.08%; $P=0.016$), whereas the DBP decline rate did not change significantly (week 12: 9.19±7.62% versus baseline: 7.78±6.92%; $P=0.176$). Furthermore, the reduction in SBP with finerenone at week 12 was persistent across all time points recorded by ABPM, with the mean ambulatory SBP curve consistently below the mean baseline values (Figure 2A). Similar trends were observed for the mean hourly DBP (Figure 2B). Office BP showed more pronounced reductions, with SBP and DBP decreasing by 15.58±1.69 and 8.61±1.02 mm Hg, respectively (both $P<0.001$ versus baseline). Office BP consistently declined from baseline through the treatment period (Figure 2C), and the proportion of patients who achieved normal office BP increased to 43.6%, 50.9%, and 61.8% at weeks 4, 8, and 12, respectively (Figure 2D).

Serum potassium increased by 0.39±0.05 mmol/L from baseline at week 12 (week 12: 4.01±0.32 mmol/L versus baseline: 3.63±0.39 mmol/L; $P<0.001$). The most substantial increase in serum potassium occurred at week 4, with levels remaining relatively stable at weeks 8 and 12 (Figure 3A). The proportion of patients with normal serum potassium levels increased progressively from 61.8% (baseline) to 81.8%, 90.9%, and 94.5% at weeks 4, 8, and 12 after treatment, respectively (Figure 3B). Patients were categorized into hypokalemic (K^+ <3.5 mmol/L) and normokalemic (K^+ ≥3.5 mmol/L) groups on the basis of their baseline levels. In the hypokalemic group, 67%, 91%, and 100% of patients achieved serum potassium increases of ≥0.5, ≥0.25, and ≥0 mmol/L, respectively, from baseline to week 12 (Figure 3C). The normokalemic group had 21%, 53%, and 76% increases for ≥0.5, ≥0.25, and ≥0 mmol/L, respectively (Figure 3D). Notably, 8 patients (24%) in this group experienced reductions in serum potassium at week 12. However, only 1 patient had a serum potassium concentration <3.5 mmol/L (K^+: 3.43 mmol/L) at this time point. This individual received potassium chloride extended-release tablet supplementation during the run-in period.

PRA remained elevated throughout the treatment period (week 12: 0.62 [0.18–1.27] ng/mL per h versus baseline: 0.23 [0.06–0.56] ng/mL per h; $P<0.001$; Figure 4A). Eighteen (32.7%) patients achieved reversal of renin suppression (PRA ≥1 ng/mL per h) at week 12. Aldosterone increased after treatment (week 12: 32.20 [23.00–51.70] ng/dL versus baseline: 25.30 [19.70–37.20] ng/dL; $P<0.001$; Figure 4B). Notably, the UACR exhibited a sustained reduction over time (week 12: 18.80 [6.30–34.70] mg/g versus baseline: 30.50 [13.90–66.33] mg/g; $P=0.002$). During the study period, the eGFR slightly decreased at week 4 and remained relatively stable at weeks 8 and 12 with no significant changes observed between week 12 and baseline (week 12: 96.38±19.75 mL/min per 1.73 m^2 versus baseline: 98.99±18.08 mL/min per 1.73 m^2; $P=0.165$; Figure 4C). One patient in this study had an eGFR of 64.3 mL/min per 1.73 m^2 at week 8 (>30% decline from baseline), which recovered to 76.6 mL/min per 1.73 m^2 by week 12.

When the standardized PAMO criteria were applied to the 55 patients in this study, 16 (29.1%) had a complete biochemical response, 24 (43.6%) had a partial biochemical response, and 15 (27.3%) had an absent response. In terms of clinical outcomes, 11 (20.0%) patients achieved a

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Table 2. Changes in Outcomes From Baseline in Study Patients During Treatment

Variables	Baseline	Week 12	Change from baseline	*P* value
24-h ABPM				
Daytime SBP, mm Hg	138.40±12.04	131.71±12.09	−6.69±1.60	<0.001
Daytime DBP, mm Hg	88.31±8.50	83.76±8.75	−4.55±1.06	<0.001
Nocturnal SBP, mm Hg	131.67±13.55	122.04±15.36	−9.64±1.60	<0.001
Nocturnal DBP, mm Hg	81.42±9.68	76.04±10.04	−5.38±1.00	<0.001
24-h mean SBP, mm Hg	136.82±12.05	129.07±12.34	−7.75±1.48	<0.001
24-h mean DBP, mm Hg	86.73±8.57	81.71±8.76	−5.02±0.95	<0.001
Nocturnal SBP decline rate, %	4.99±6.08	7.14±6.46	2.15±0.87	0.016
Nocturnal DBP decline rate, %	7.78±6.92	9.19±7.62	1.41±1.03	0.176
Office SBP, mm Hg	142.73±11.43	127.15±12.73	−15.58±1.69	<0.001
Office DBP, mm Hg	95.88±8.15	87.27±8.23	−8.61±1.02	<0.001
Serum potassium, mmol/L	3.63±0.39	4.01±0.32	0.39±0.05	<0.001
eGFR, mL/min per 1.73 m^2	98.99±18.08	96.38±19.75	−2.61±1.85	0.165
PRA, ng/mL per h	0.23 (0.06 to 0.56)	0.62 (0.18 to 1.27)	0.17 (0 to 0.86)	<0.001
PRA ≥1.0 ng/mL per h, n (%)		18 (32.7)		
ALD, ng/dL	25.30 (19.70 to 37.20)	32.20 (23.00 to 51.70)	8.80 (1.50 to 17.80)	<0.001
UACR, mg/g	30.50 (13.90 to 66.33)	18.80 (6.30 to 34.70)	−8.30 (−18.80 to 0.60)	0.002

ABPM indicates ambulatory blood pressure monitoring; ALD, aldosterone; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; PRA, plasma renin activity; SBP, systolic blood pressure; and UACR, urinary albumin-to-creatinine ratio.

complete clinical response, 22 (40.0%) achieved a partial clinical response, and 22 (40.0%) had an absent response (Figure 4D). Overall, 5 (9.1%) patients achieved a complete biochemical and clinical response to finerenone treatment.

A sensitivity analysis was performed that included 2 patients who were excluded from the primary analysis because of study withdrawal. Both analysis sets (N=57 and N=55) showed consistent treatment efficacy outcomes, supporting the validity of the primary analysis.

Safety

This study reported a low incidence of AEs, with only 2 (3.5%) patients experiencing such events post-treatment. One patient reported a persistent mild headache during the treatment period, whereas the other sustained a minor fracture due to trauma. No sex hormone receptor–related AEs or hyperkalemia were observed throughout the study.

DISCUSSION

Currently, the pharmacological management of PA remains challenging because of the limited options, unavoidable side effects, and inadequacy of the treatment response.[15] Finerenone, a novel nonsteroidal MRA with superior selectivity and favorable tolerability, is primarily indicated to slow the progression of CKD in patients with type 2 diabetes and improve cardiovascular outcomes.[19] Emerging but limited evidence has evaluated its efficacy in patients with PA.[21,22,25,26] In this prospective, multicenter, single-arm, and exploratory study, we found that finerenone is associated with a reduced severity of hypertension, relieved hypokalemia, and increased renin levels in patients with PA without inducing obvious side effects. These findings add novel evidence that finerenone can be used as a potential alternative medication to present strategies for PA treatment.

Prior evidence for the therapeutic effect of finerenone in PA was limited.[25,26] In a recently published real-world PA study, after 15 patients were switched from eplerenone to finerenone and with a median follow-up of 5 months, BP and antihypertensive medication doses remained stable. Notably, the proportions of patients with normalized BP and biochemical remission declined significantly.[22] To date, the only randomized controlled trial that evaluated finerenone treatment for PA showed comparable antihypertensive efficacy and no sex-related adverse effects compared with spironolactone.[21] However, this was a single-center study with an 8-week observation period. In contrast, our study is the first multicenter study to extend the duration to 12 weeks. Furthermore, in our protocol, patients were maintained on a standardized stable-dose nifedipine regimen throughout the trial to minimize confounding factors from heterogeneous concomitant medications. The above measures increase the solidity of this study.

Previous studies that have assessed MRA efficacy in PA have relied primarily on office BP measurements.[27] Compared with office BP, daytime SBP measured by 24-hour ABPM is more strongly correlated with cardiovascular disease, stroke risk, and all-cause mortality.[28–30] Therefore, both our study and Hu et al[21] adopted mean daytime SBP changes as the primary outcome.

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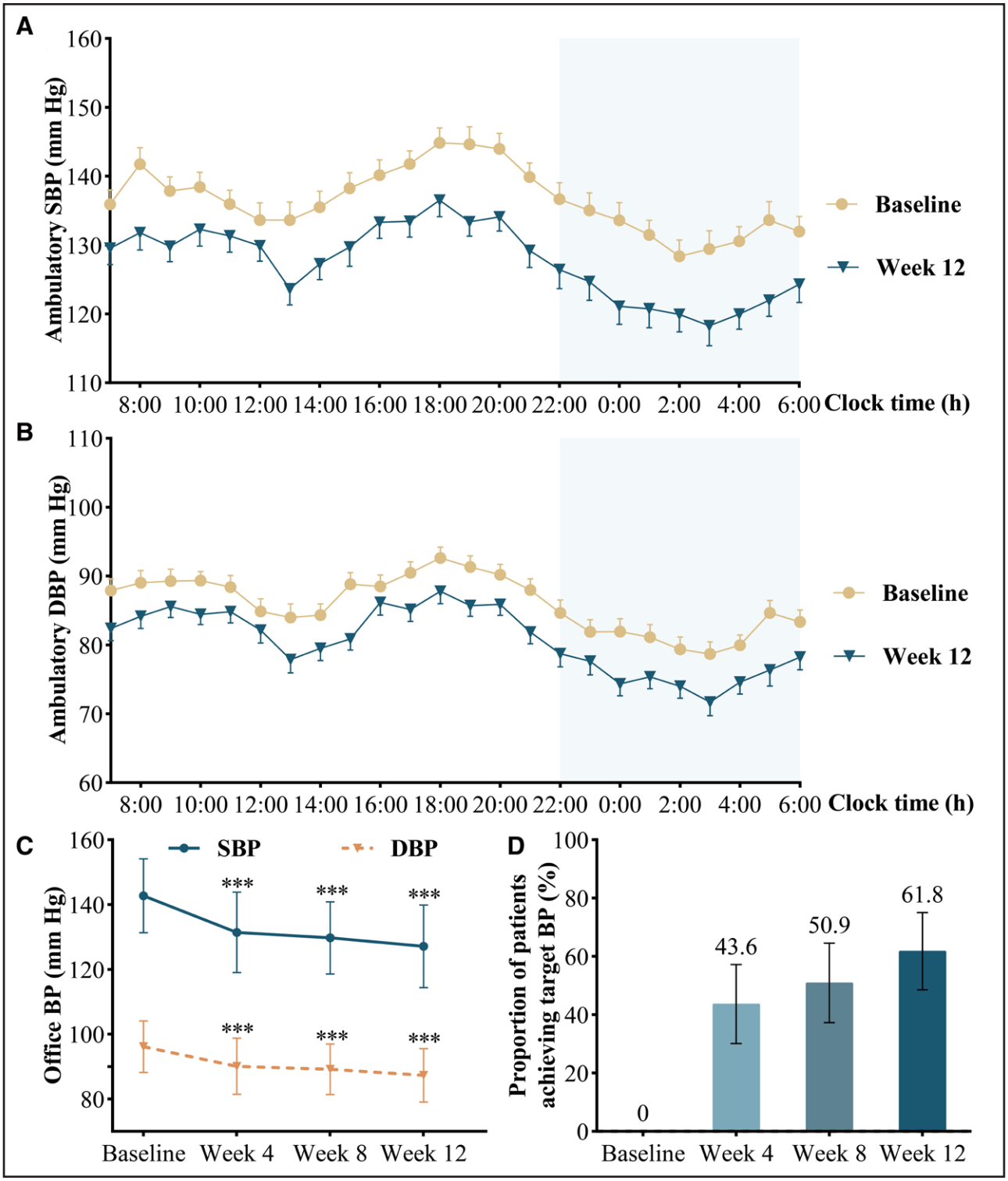

Figure 2. Changes in 24-hour ambulatory blood pressure (BP) and office BP during treatment.
Change in systolic blood pressure (SBP; **A**) and diastolic blood pressure (DBP; **B**) over 24 hours recorded by ambulatory BP monitoring; change in office BP over time (**C**); and proportion of patients who achieved the target BP over time (BP <140/90 mm Hg; **D**). The error bars represent the means with SEs (**A** and **B**), SDs (**C**), and 95% CIs (**D**). ***P<0.001 vs baseline.

Consistent with Hu et al's findings, we also observed significant reductions in ambulatory BP. Notably, we discovered that nocturnal BP reductions were more pronounced than daytime reductions. Aldosterone-mediated sodium and water retention blunts the normal nocturnal BP decline. The improved nocturnal BP reduction in our study may indicate the correction of PA pathology by finerenone.[31,32] Moreover, both studies reported

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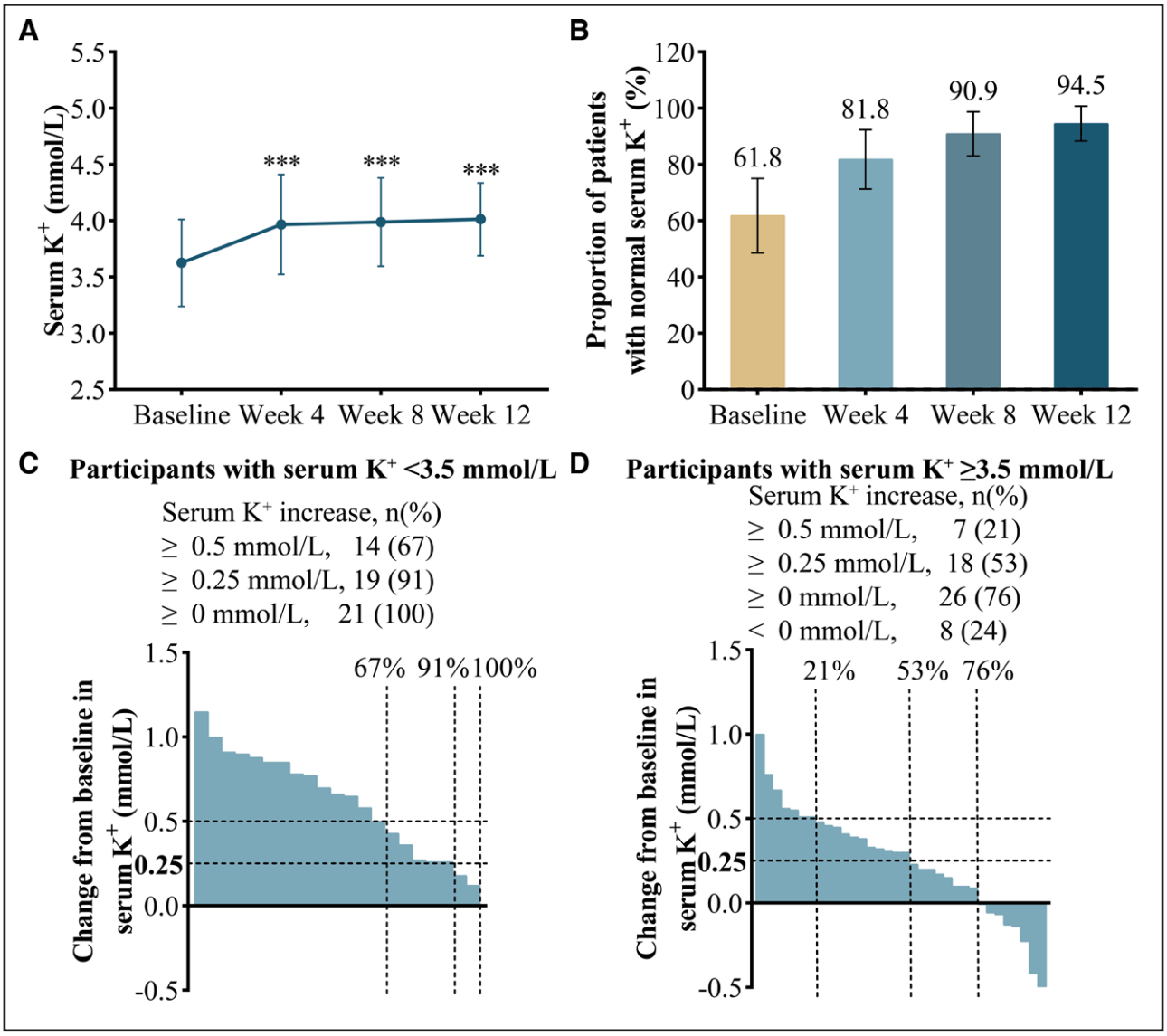

Figure 3. Changes in serum potassium levels during treatment.
Changes in serum potassium levels over time (**A**); proportions of patients who achieved normal serum potassium levels (3.5–5.5 mmol/L) over time (**B**); and individual changes in serum potassium levels from baseline to week 12 in the subgroups of patients with serum potassium levels <3.5 mmol/L (**C**) and ≥3.5 mmol/L (**D**). The error bars represent the means with SDs (**A**) and 95% CIs (**B**). ***P<0.001 vs baseline.

significant improvements in office BP (Hu et al: −17.7/−9.1 mm Hg versus ours: −15.58/−8.61 mm Hg) and similar BP target achievement rates (Hu et al: 66.7% versus ours: 61.8%). In addition, our study revealed progressively greater BP reductions and higher BP control rates over extended treatment durations.

In the ARTS-DN ABPM substudy, Agarwal et al[33] demonstrated that finerenone (10–20 mg/d) reduced the 24-hour mean SBP by 8.3 to 11.2 mm Hg in patients with CKD with type 2 diabetes. However, 2 pivotal phase III clinical trials (FIDELIO and FIGARO)[18–20] revealed that finerenone reduced the mean office SBP by only 3.7 to 5.74 mm Hg in patients with CKD with type 2 diabetes. This magnitude was notably smaller than the reductions observed in our and Hu et al's PA populations.[21] Although a larger maximum dose (40 mg/d) of finerenone was used in our study and Hu's study, these differential responses strongly suggest that finerenone's direct antagonism of mineralocorticoid receptors confers superior antihypertensive efficacy in patients with PA. Our study was not designed to directly compare the antihypertensive efficacy of finerenone and spironolactone. The findings from Hu et al suggest comparable BP-lowering effects between finerenone and a lower dose of spironolactone in patients with PA. A real-world study suggested that finerenone has comparable antihypertensive efficacy to eplerenone.[22] Previous studies focusing on esaxerenone (another nonsteroidal MRA) demonstrated similar antihypertensive efficacy to that of finerenone in our study.[24,34] However, this comparison was determined to be indirect.

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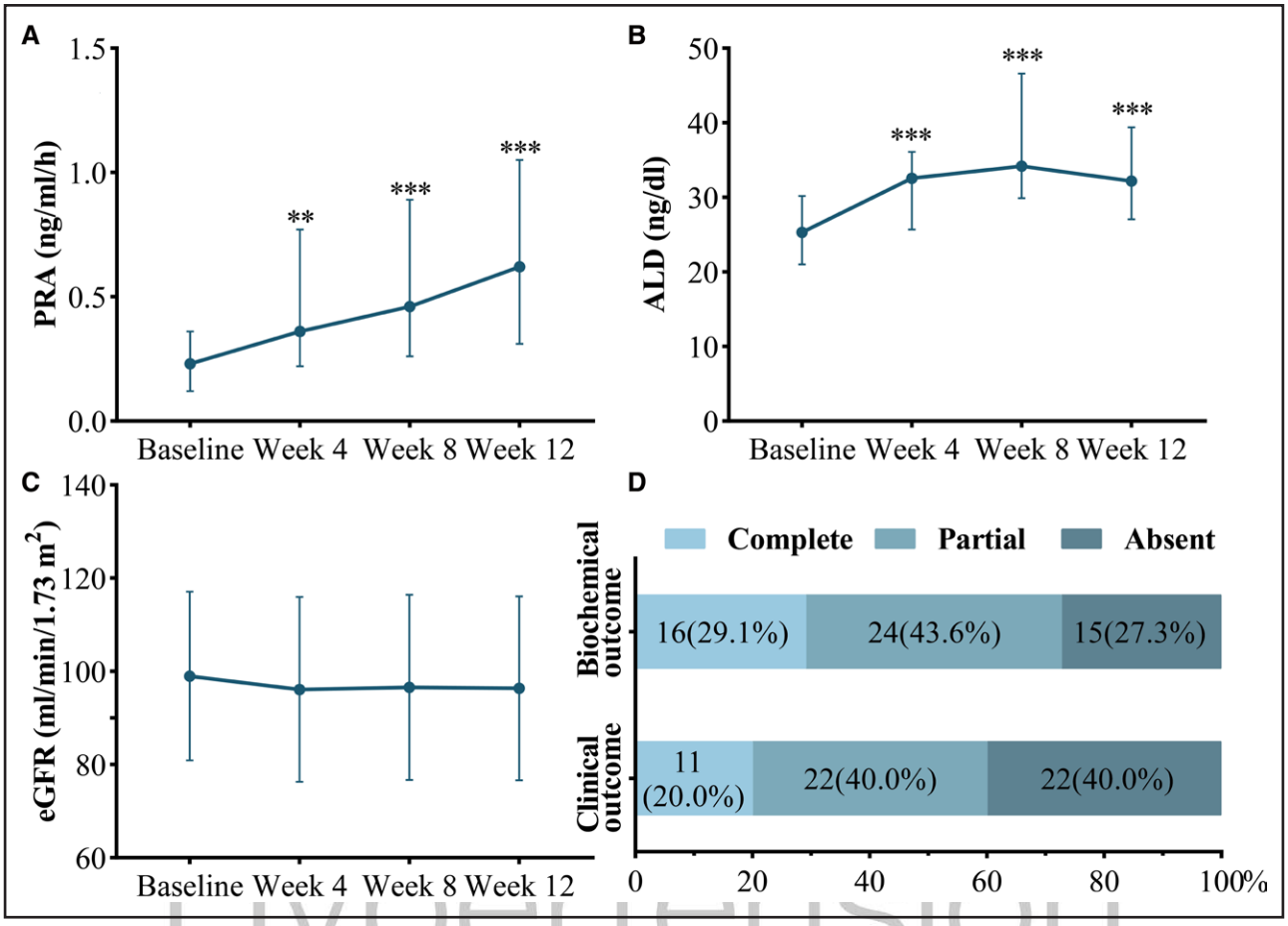

Figure 4. Changes in pharmacodynamic measures and treatment outcomes.
Changes in plasma renin activity (PRA; **A**), aldosterone (ALD; **B**), and estimated glomerular filtration rate (eGFR; **C**) levels over time; counts and proportions of patients in different biochemical and clinical outcome groups after 12 weeks of treatment (**D**). The error bars show the medians with 95% CIs (**A** and **B**) and the means with SDs (**C**). $^{**}P<0.01$ and $^{***}P<0.001$ vs baseline.

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Future randomized controlled trials are needed to compare the efficacy and safety of finerenone with those of other MRAs in patients with PA.

The normalization of serum potassium also serves as a biomarker for medical therapy in PA.[15] Our findings demonstrate that finerenone treatment promotes the resolution of hypokalemia in patients with PA. The increase in potassium was more pronounced in patients with hypokalemia at baseline, and no patients experienced a decrease in serum potassium in this subgroup. Notably, a prominent increase in serum potassium was observed at 4 weeks post-treatment, and almost 95% of patients achieved normal potassium levels at the end of treatment. In addition, our study revealed greater serum potassium elevation (0.39 versus 0.2 mmol/L) than Hu et al's findings.[21] This difference may be attributed to lower baseline potassium levels, longer treatment durations, and higher average finerenone doses in our cohort.

The normalization of BP and serum potassium levels does not reflect adequate aldosterone blockade at the tissue level.[15] As indicated in the newly published PAMO study, elevated renin levels (from a suppressed state to normalization, PRA ≥1.0 ng/mL per h) are significantly associated with improvements in cardiovascular and renal outcomes and serve as a key marker for the reversal of PA pathophysiology.[15] Consistent with the findings of Hu et al,[21] our study revealed an increase in PRA levels in patients with PA after finerenone treatment, with 32.7% of patients achieving PRA ≥1.0 ng/mL per h. These findings confirm that finerenone reverses the underlying pathophysiology in a subset of patients with PA. We further assessed the biochemical and clinical response rates of finerenone-treated patients on the basis of the PAMO criteria.[15] In this study, the complete biochemical response rate with finerenone was 29.1%, which was higher than that reported in a real-world study of PA (11%).[22] This difference may be attributed to the higher average dosage used in our study (30 mg versus 20 mg in the real-world study). The real-world data also demonstrated lower biochemical response rates with finerenone than with eplerenone. The PAMO study investigated the outcomes of PA with the present treatment (mainly spironolactone) in an international cohort and revealed a higher complete biochemical response

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rate (52.9% versus our 29.1%) and a similar complete clinical response rate (18.3% versus our 20.0%) compared with our study. Although current therapies appear to have higher biochemical response rates than finerenone does, this does not sufficiently indicate that finerenone is less effective than current MRAs are. The main reasons are given as follows: (1) compared with the minimum of 6 months in the PAMO research, the 3-month observation period in our study was relatively short to assess therapeutic efficacy; (2) the maximum effective dosage of finerenone for treating PA remains undefined; notably, traditional MRA spironolactone can be prescribed up to 400 mg and has dose-dependent efficacy; and it is reasonable to infer that higher doses of finerenone might produce better outcomes; and (3) unlimited and various background drugs in addition to finerenone may have influenced the biochemical response rates in the PAMO study.

In addition, we observed that finerenone induced a significant reduction in UACR among patients with PA, which was consistent with previous findings about patients with CKD with type 2 diabetes and suggested that finerenone also leads to renal benefits for patients with PA.[18,20,35] Notably, in our cohort, only 50% of participants exhibited abnormal UACR levels with mild elevation. Future studies evaluating renal outcomes in patients with PA are needed to clarify the nephroprotective potential of finerenone in patients with PA. In direct comparison of preclinical models of cardiorenal injury, finerenone conferred more potent anti-inflammatory and antifibrotic effects than steroidal MRAs, thereby plausibly reflecting its more complete blockade of mineralocorticoid receptor–driven transcription.[36–39] Furthermore, finerenone exhibits a balanced distribution in the heart and kidneys, and it has been demonstrated to exhibit cardiorenal protective effects in patients with CKD with type 2 diabetes.[18,20,40] Thus, it is reasonable to propose that finerenone may offer greater target-organ-protection potential in PA although further investigations are needed to confirm this notion.

During the study, we observed a transient decline in the eGFR following treatment initiation that recovered during long-term therapy. The initial decrease in the eGFR likely reflects the correction of aldosterone-mediated glomerular hyperfiltration, suggesting that the restoration of tubuloglomerular feedback is impaired by chronic hyperaldosteronism.[41–43] Consistent with the findings reported by Hu et al,[21] no significant AEs (including sex hormone-related adverse effects) were observed in our study. While hyperkalemia has been observed in patients with CKD with type 2 diabetes who receive finerenone treatment, no such cases were identified in either our cohort or Hu et al's PA cohort. This was likely due to the inherent hypokalemic tendency in patients with PA. Our study confirms the safety of finerenone 40 mg/d in patients with PA with an eGFR ≥60 mL/min per 1.73 m². The maximum effective and safe dosage of finerenone for patients with PA requires further determination. The incidence of hyperkalemia in patients with PA with an eGFR <60 mL/min per 1.73 m² receiving finerenone treatment remains unclear and warrants further investigation.

In addition to MRAs, ASIs remain under active investigation for PA.[14] A recently published study demonstrated the promising therapeutic potential of this drug class for PA management. A phase 2 clinical trial of baxdrostat (an ASI) indicated effective antihypertensive efficacy (−24.9/−10.6 mm Hg).[14] In addition, baxdrostat significantly reduced aldosterone levels, increased renin levels, and corrected hypokalemia. Safety findings suggested that no AEs occurred within the first 12 weeks of treatment; nevertheless, hyperkalemia and renal function decline occurred thereafter.[14] Large-scale randomized controlled trials are still needed to confirm the efficacy and long-term safety of this treatment. Currently, no direct evidence has compared the use of baxdrostat with finerenone in the context of PA treatment. Given that ASIs are also devoid of gonad-related side effects, whether ASIs can replace MRAs as a first-line therapy for PA or whether combination therapy is viable requires further investigation.

Several limitations warrant consideration in this study. First, the single-arm design lacked a placebo or an active comparator, which weakened the efficacy of the evaluation. Second, the exclusion of patients with an eGFR <60 mL/min per 1.73 m² restricts the generalizability of our findings to all patients with PA. Third, while the 12-week observation period was adequate for a preliminary evaluation of efficacy, long-term follow-up is needed to fully assess the sustained therapeutic effects and safety. Finally, subgroup analyses were not performed in this study. The current evidence remains insufficient to identify patients who would derive maximum therapeutic benefit from finerenone treatment.

In conclusion, treatment with finerenone is associated with a reduced severity of hypertension, relieved hypokalemia, and increased renin levels in patients with PA without inducing obvious side effects. These findings support the use of finerenone as a potential alternative therapy for PA. Before considering finerenone as a substitute for spironolactone, further comparative and outcome studies with longer follow-up durations are warranted to verify the favorability of the profiles of finerenone in patients with PA.

PERSPECTIVES

Current pharmacotherapy for PA remains suboptimal, emerging ASIs are still in the clinical research phase, and existing MRAs have unavoidable side effects and lead to inadequate treatment response. These limitations highlight the need for better therapeutic alternatives for PA. The novel nonsteroidal MRA finerenone exhibits superior receptor selectivity and offers definitive cardiorenal

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protective benefits in patients with CKD with type 2 diabetes; however, its clinical efficacy and safety in the context of treating PA have not been fully determined. Our study confirms the efficacy and safety of finerenone in patients with PA, thus suggesting that finerenone is a potential alternative therapy for PA. Future large-scale placebo-controlled and standard-of-care arms featuring longer follow-up durations are essential to provide robust evidence to support the use of finerenone for PA treatment. Furthermore, finerenone may offer unique advantages with respect to reducing the risk of cardiovascular and renal damage in patients with PA although further investigations are required to confirm this claim.

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Disclosures

None.

Supplemental Material

Tables S1–S2

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